Siddhanth Yellanki

ee24btech11059@iith.ac.in | siddhanthyellanki.07@gmail.com | +91 7893883341 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.34
Narayana Junior College	Class XII (SSC)	2024	93.80%
Bharatiya Vidya Bhavan Public School	Class X (CBSE)	2022	90.40%

SKILLS

Languages: C, C++, Embedded C, AVR Assembly, Python, Verilog

ML & Deep Learning: PyTorch, CNN, Deep Learning, Model Compression, Quantization, Pruning, NumPy, Scikit-learn, Evolutionary Search (NSGA-II), GPU-Accelerated Training

Computer Vision & Signal Processing: OpenCV, Image Processing, Signal Processing, Depth Camera / Point Cloud Processing, ROS2 TF, Hough Transform

Robotics & Simulation: ROS2, Nav2, SLAM Toolbox, MoveIt2, Gazebo, ArduPilot, MAVLink, Arduino IDE

Embedded & Hardware: STM32, ESP32, Raspberry Pi, Pixhawk 2.4.8, AVR Microcontrollers, FPGA (Verilog)

Developer Tools: Git, Linux, VS Code, PlatformIO, Mission Planner

ACHIEVEMENTS

e-Yantra Robotics Competition (EYRC) | *IIT Bombay* -- Jun 2025 - Dec 2025

- Qualified and advanced to **Stage 2** of the national robotics competition organised by IIT Bombay, competing against over 400 teams and 34,500 students from across India.
- Worked on real-time robotics tasks involving **perception**, **autonomous navigation**, and **robotic manipulation** under strict competition constraints and deadlines.

PROJECTS

Intelligent 6-DOF Robotic Manipulator

- **Implemented** an automated **image processing** and sorting pipeline using a 6-DOF arm; **applied** OpenCV color-space thresholding and texture analysis (**signal processing**) to identify rotten fruit, with 3D coordinates from depth-camera point cloud processing.
- **Solved** inverse kinematics constraints using **MoveIt2** for smooth path planning and collision avoidance in a cluttered simulation environment.

Holonomic Station-Seeker

- **Developed** an **Artificial Potential Field** (APF) controller for a 4-wheel holonomic robot; **implemented** repulsive field gradients (**signal processing**) for obstacle navigation and goal-directed attraction.
- **Processed** raw LIDAR sensor data using a Hough Line Transform (**signal processing** algorithm) to extract geometric station signatures, enabling precise docking in **noisy environments**.

Neural Architecture Search - Evolutionary Search with Pareto-Optimal Selection

- **Designed and implemented** an evolutionary **deep learning** NAS pipeline (regularized evolution + NSGA-II Pareto selection) to search a **CNN**/**RegNe**t-style architecture space (width, depth, kernel, group) using **early-stopping proxy training** for efficient candidate ranking.
- **Built** a custom **accuracy/FLOPs scoring** criterion, **trained** an **SGDRegressor** surrogate model to accelerate search, and **ran** full **GPU-accelerated** convergence validation against a hand-designed **RegNetY CNN** baseline on CIFAR-10.
- **Achieved 87% parameter reduction** for a 3% accuracy tradeoff, and another **matching baseline accuracy (+0.4%) with 38% fewer parameters** - demonstrating **model compression** and efficiency optimization.

Per-Layer Precision & Sparsity Search -- MXFP Quantization (CIFAR-10)

- **Built** a greedy per-layer precision search over block-scaled **MXFP** formats for a **CNN-based ResNet,** wrapping Conv2d/Linear layers with a **runtime-switchable quantizer**, combined with magnitude pruning for **model compression**.
- **Ran quantization-aware finetuning** (deep learning training pipeline) to recover accuracy, then **benchmarked** FP32 baseline vs MXFP-only vs pruning-only using an accuracy-drop tolerance as selection criterion.
- **Achieved ~93% storage-bit reduction** via mixed per-layer MXFP formats within a 5% accuracy tolerance -- direct **model compression** and **performance optimization.**

POSITIONS OF RESPONSIBILITY

Mechatronics Coordinator -- Jan 2025 - May 2026 -- Tinkerer's Lab, IITH, Telangana, India

- Conducted **3 hands-on robotics workshops** for school students (average of 40 attendees each); designed session curriculum covering actuator selection, basic kinematics, and microcontroller interfacing.
- Built a **flight-capable quadcopter** end-to-end: frame assembly, motor-ESC wiring, power distribution, and Pixhawk flight controller tuning.
- Completed **CAD and motor selection** for a 1:1 scale Wall-E model; performed torque and load calculations to validate actuator choices for drive and articulation joints.

RELEVANT COURSES

Data Structures and Algorithms, Pattern Recognition and Machine Learning, Digital Signal Processing, Digital Image Processing, Control Systems, Semiconductor Device Fundamentals, Linear Algebra, Probability, Random Processes, Intro to Statistics, Numerical Analysis,

EXTRACURRICULAR

- Inter IIT Swimming Team, competed at national inter-institute level, balancing rigorous training with academics.
- Part of Vibes the Music Club of IITH.